

Phase 3 - MICE Imputation & Rubin's Rules Valuation

Phase 3 closes the missing-data gap in the dataset and produces the headline national valuation. Two variables require imputation: the polygon-derived acreage (`osm_acreage`, 28.8% missing) and the economic baseline value (`Baseline_Value_Per_Acre`, 6.7% missing). The imputation method is Multiple Imputation by Chained Equations (MICE) with a Random Forest backend, and the resulting per-dataset estimates are pooled via Rubin's Rules to produce a national aggregate with a defensible confidence interval.

The pipeline runs MICE independently in **Python** (`miceforest` with a LightGBM gradient-boosted Random Forest), **R** (`mice` with `method = "rf"`), and **Julia** (`Mice.jl`). Each language produces $M = 100$ complete imputed datasets, yielding 300 imputed datasets in total across the tri-language design. I chose $M = 100$ over the $M = 5$ to $M = 10$ that early MICE literature recommended because the higher value follows modern guidance for stabilizing standard errors and eliminating Monte Carlo error, and is particularly important when the missing-data rate is as high as 28.8%. I preferred Random Forest over linear MICE backends because the relationship between course acreage and its predictors (number of holes, ownership type, urban/rural classification, latitude and longitude) is non-linear, and tree-based methods cannot produce impossible negative-valued imputations the way linear methods can.

Within each language, per-dataset opportunity costs are computed as `osm_acreage` × `Baseline_Value_Per_Acre` summed across courses, then pooled across the 100 imputations using Rubin's Rules: the pooled point estimate is the mean of the per-dataset sums, and the total variance combines within-imputation variance (the dispersion of estimates within a single completed dataset) and between-imputation variance (the dispersion of point estimates across the 100 datasets). The 95% confidence interval is the pooled estimate plus or minus 1.96 standard errors.

Key result

All three languages converge on an aggregate national opportunity cost in the **\$926B–\$959B range**, with overlapping 95% confidence intervals: Python at \$943.0B (CI: \$936.3B–\$949.7B), R at \$936.0B (CI: \$926.4B–\$945.7B), and Julia at \$951.4B (CI: \$943.8B–\$958.9B). The Grand Mean across the three implementations is approximately **\$943 billion**. The cross-language spread of 1.6% on a ~\$940B base is well within expected variation from independent Random Forest RNG seeds and internal MICE implementations, and is materially smaller than the natural between-imputation variance within any single language.

The diagnostic that between-imputation variance (V_B) is two to three orders of magnitude larger than within-imputation variance (V_W) confirms the model is well-specified: uncertainty in the aggregate estimate is driven by the genuine ambiguity of the missing values, not by measurement noise within any single completed dataset.

What was solved

The Julia implementation initially failed with world-age errors when `Mice.jl` functions were defined and called in the same top-level script execution context. I refactored `Phase_3.jl` to use `Mice.jl`'s `mice()` and `complete()` APIs in the documented order, with imputation logic isolated in its own function scope. This fix was load-bearing for the cross-language validation: without a working Julia pipeline, the tri-language design would have collapsed to a Python-vs-R comparison, which would not have demonstrated robustness against backend-specific MICE implementation choices.

Phase 3 also produces a national acreage summary aggregating the imputed footprint by urban/rural classification. The pooled total is approximately 2.30 million acres, with 1.70 million in urban counties and 600,000 in rural counties. This summary is a sanity check on imputation stability rather than a primary research output, and feeds directly into the regression and visualization work in Phases 4 and 6.

Outputs flow forward into Phase 4 (regression analysis and marginal effects estimation) and ultimately into Phase 6 (visualization and table generation for the thesis).